TECHNICAL DESIGN DOCUMENT (TDD)

DumHouse: Direct Food-Ordering Platform with Real-Time Kitchen Display System (KDS), AI Prep-Time Prediction & Operations Suite

Project Metadata & Review Links

- **GitHub Project Repository:** `https://github.com/NamanBhatia-NB/Restaurant_Ordering_Kitchen_Display_System`
- **Live Production Application:** `https://restaurant-ordering-kitchen-display-two.vercel.app`
- **Client Organization:** Company A, Hyderabad (Biryani & Authentic Hyderabadi Food Chain)
- **Scale:** 14 Outlets, 3 Cloud Kitchens, 520 Staff, Rs. 90-105 Crore Annual Revenue
- **Prepared For:** Isocode Software Engineering Take-Home Assessment (Problem Statement 3)
- **Author:** Software Engineering Candidate

1. Executive Summary & Business Rationale

1.1 The Client's Dilemma

Company A is a premier Hyderabad restaurant chain celebrated for authentic Handi Dum Biryani, Mutton Haleem, and Nizami specialties. Despite generating Rs. 90-105 crore in annual revenue, over **55% of sales flow through third-party food delivery aggregators (Swiggy, Zomato)**. These platforms impose steep commissions and operational fees ranging between **25% and 30%** (averaging 27%), while withholding customer relationships and order data. Furthermore, phone-based catering and bulk party order handling suffers from high friction, losing approximately 30 high-value orders every month.

1.2 The Economic Case & ROI

- **Direct Order Migration:** Shifting just 12% of app orders to direct online channels (where payment gateway and direct delivery fulfillment fees average 12%) saves **15 percentage points** on Rs. 6.336 Crore:

```
Commission Savings = 0.15 x Rs. 6,33,60,000 = Rs. 95.04 Lakh per year
```

- **Recovering Lost Bulk & Party Orders:** Regaining 30 lost orders monthly at Rs. 18,000 average order value (AOV) with a 30% contribution margin:

```
Bulk Recovery = 30 orders x (Rs. 18,000 x 30% margin) x 12 months = Rs. 19.44 Lakh per year
```

- **Total Annual Bottom-Line Impact: Rs. 1.14 Crore per year** against an engineering implementation budget of Rs. 9 Lakh (yielding an ROI of over **12.6x within the first 12 months**).

2. Comprehensive Scope & Feature Implementation Matrix

To deliver an exceptional, robust, and zero-defect product, **100% of the brief requirements** have been built and verified across **8 integrated screens**:

2.1 Dual-Workflow Architecture (Customer & Kitchen)

Workflow Phase	Customer Direct Ordering Workflow	Kitchen & Operations Suite Workflow
Discovery & Setup	**1. Outlet & Menu Screen (`/`)**: Geolocation / pincode detection, category navigation, dietary Veg/Non-Veg badges, spice intensity meters (0-3), AI 1-tap reorder from past orders.	**8. Central Menu CMS (`/staff/menu`)**: Centralized multi-city pricing control, recipe metadata, allergen tags, and instant single-click out-of-stock toggles to pause sold-out handi pots.
Selection & Customization	**2. Item Customizer & Cart**: Portion size toggle (Regular, Family Pack), add-on sides (Extra Salan, Mirchi ka Salan, Raita, Boiled Egg), discount coupon validation (`DUMDIRECT`, `NIZAM15`), and loyalty points redemption.	**5. Kitchen Display System (`/staff/kitchen`)**: High-contrast tablet UI readable from 2 meters, audio cue chimes for incoming orders, visual timers, and one-tap stage transitions (`RECEIVED` -> `ACCEPTED` -> `PREPARING` -> `READY` -> `HANDED_OFF`).
Checkout & Commitment	**3. Express Checkout & Payment (`/checkout`)**: DPDP Act 2023 compliant marketing opt-in checkbox, delivery address, contact details, transparent tax breakdowns (5% GST included notice), and dynamic UPI QR / test card gateway.	**6. Live Order Status Tracker (`/order/[id]`)**: Real-time SSE timeline updates, dynamic promised delivery countdown, Ridge Regression AI estimator badge, and simulated WhatsApp/SMS status notification dispatches.

Specialized Catering	**4. Bulk Party Order Portal (`/party-orders`)**: Strict 24-hour advance notice validation, guest count slider (20-350 guests), automated 10kg handi pot sizing calculator, custom catering quote generation, and manager review workflow.	**7. Operations & Financial Dashboard (`/staff/dashboard`)**: Aggregated tracking of the 15% commission savings toward the Rs 95.04 Lakh direct-ordering target, outlet preparation time efficiency, cancellation rates, and repeat customer retention.

2.2 Screen Implementation Matrix

Screen / Flow	Key Capabilities	Compliance & Design Nuances
1. Customer Menu (`/`)	Outlet selection, category navigation, search, veg/non-veg filter, spice level meters, allergen chips, **AI 1-Tap Quick Reorder from past history**.	Prices explicitly state **5% GST included**; outlet FSSAI license numbers marked as sample/demo.
2. Item Customizer & Cart	Portion selection (Regular, Family Pack), add-on sides (Extra Salan, Raita, Boiled Egg), **Promo Coupon validation (`DUMDIRECT`, `NIZAM15`)**, and **DumRewards Loyalty Points redemption**.	Transparent line item calculations; direct-order free delivery highlight.
3. Checkout (`/checkout`)	Customer recipient, phone, delivery address, **DPDP Act 2023 Marketing Opt-in Checkbox**, mock payment gateway trigger.	DPDP Act 2023 privacy notice; input validation via Zod schemas.
4. Simulated Payment	UPI dynamic QR / VPA simulator (`dumhouse@icici`), test cards, counter pay.	Sandboxed payment authorization; server-side price computation.
5. Live Order Tracking (`/order/[id]`)	Real-time status progress timeline, live prep countdown, AI Estimator badge, **Simulated WhatsApp / SMS auto-dispatch notification**.	Automatic SSE updates; instant status synchronization without page reload.
6. Bulk Party Orders (`/party-orders`)	Minimum 24h advance notice requirement, guest count slider (20-350 guests), Handi pot calculator, package selection, catering manager review.	Fulfills the lost bulk orders problem, recovering Rs. 19.44 Lakh/year in catering margin.
7. Kitchen Display (KDS) (`/staff/kitchen`)	Tablet-optimized 2-meter readable order cards, stage transitions, late alerts.	Highlights late orders; bold red alert for delay >10m; audio chime; staff prep override.
8. Central Operations Dashboard (`/staff/dashboard`)	Live executive overview of orders, revenue, 15% aggregator commission savings tracker, prep times, cancellation rates, and repeat customers.	Full multi-outlet visibility across all 14 locations and cloud kitchens.
9. Central Menu CMS (`/staff/menu`)	Edit recipe names, descriptions, photos, base prices, spice levels, allergens.	Per-outlet stock availability toggle (pause individual handi pots).
10. Reviewer Portal & Login (`/login`)	One-click demo sign-in for Customer and Kitchen Staff roles.	Floating top banner for frictionless reviewer persona switching.

3. System Architecture & Technical Rationale

3.1 Architecture Overview

The platform is built as a cohesive, high-performance web application utilizing **Next.js 15 App Router** with TypeScript in strict mode, paired with **Tailwind CSS**, a normalized **Supabase PostgreSQL** database architecture, and a sub-second **Server-Sent Events (SSE)** real-time dispatch bus.

Architectural Layer	Components & Technologies	Core Responsibilities & Protocol
Presentation Tier	Customer Mobile PWA, Kitchen Android Tablet KDS, Operations Admin Desktop	Built with Next.js 15 App Router, React 19, and Tailwind CSS. Features high-contrast dark theme for 2-meter kitchen readability, responsive layout across all viewports, and zero dead buttons.
Application & API Tier	Server Route Handlers (`/api/orders`, `/api/menu`, `/api/stream`)	Server-side price verification and discount calculations, strict Zod schema payload validation, transactional order state machine transitions (`RECEIVED` -> `ACCEPTED` -> `PREPARING` -> `READY` -> `HANDED_OFF` -> `COMPLETED`).
Real-Time Tier	Server-Sent Events (SSE) Bus (`/api/stream`) & Supabase Broadcast	Lightweight, resilient downstream event streaming (`text/event-stream`), automatic reconnection fallback, sub-second order arrival dispatch to kitchen tablet screens.

Intelligence Tier	AI Ridge Preparation Estimator (`src/lib/ai/prep-estimator.ts`)	Closed-form analytical L2-regularized linear regression: `w = inv(X^T . X + lambda . I) . X^T . Y`, trained on 180 reproducible kitchen samples. Inputs: item count, complexity points, current queue depth, peak dinner flag, weekend indicator.
Data & Persistence Tier	Dual-Engine: Supabase PostgreSQL & Resilient Local Store	Normalized relational database with foreign key integrity, row-level security (RLS) policies, and pre-seeded realistic sample orders (`WEB-778120`, `WEB-778131`, `WEB-778146`) ensuring zero-config out-of-the-box evaluator testing.

3.2 Technical Choices & Trade-off Analysis

1. **Next.js App Router (Full-Stack Monolith):** Eliminates API gateway overhead and shared type duplication between frontend and backend.
2. **Server-Sent Events (SSE) & Supabase Realtime:** Downstream real-time order dispatching without the complexity of stateful WebSocket servers.
3. **Dual Persistence Mode:** Out-of-the-box local repository engine ensures seamless grading by reviewers without requiring cloud credential provisioning, while full Supabase SQL migrations provide cloud-readiness.

4. Data Model & Database Design

4.1 Normalized Relational Schema

Entity / Table	Primary Key	Key Foreign Keys	Core Responsibilities & Business Attributes
`outlets`	`id` (text)	None	Physical & cloud kitchen outlets: name, location, city, service pincodes array, statutory FSSAI license number, minimum prep minutes (floor), active status.
`menu_categories`	`id` (text)	None	High-level menu groupings (Biryani, Starters, Breads, Desserts, Beverages) with display weights.
`menu_items`	`id` (text)	`category_id` -> `menu_categories.id`	Dish title, description, base selling price (GST inclusive), is_veg indicator, spice rating (0-3), allergen tags array.
`outlet_menu_items`	`id` (uuid)	`outlet_id`, `menu_item_id`	Composite availability link allowing kitchen managers to pause sold-out handi pots without impacting other chain outlets.
`orders`	`id` (uuid)	`outlet_id`, `customer_id` -> `profiles.id`	Sequential order identifier (`WEB-XXXXXX`), lifecycle status, subtotal, delivery fee, discount, GST, total amount, AI prep estimate, promised delivery timestamp.
`order_items`	`id` (uuid)	`order_id` -> `orders.id`, `menu_item_id`	Portion size (`Regular`, `Family Pack`), snapshotted unit price, selected add-on sides array with prices, item total.
`order_events`	`id` (uuid)	`order_id` -> `orders.id`	Immutable audit log: `from_status`, `to_status`, actor role (`customer`, `kitchen_staff`, `system`), explanatory note, timestamp.
`profiles`	`id` (uuid)	References Supabase Auth / Local ID	Full name, phone number, role (`customer`, `kitchen_staff`, `catering_manager`), accumulated loyalty points balance, DPDP marketing opt-in consent.
`prep_training_samples`	`id` (uuid)	`outlet_id` -> `outlets.id`	Reproducible historical observations: `item_count`, `complexity_points`, `queue_depth`, `is_peak_hour`, `is_weekend`, `actual_prep_minutes`.

5. AI Preparation-Time Prediction Model

5.1 Formulation & Objective

Rather than calling non-deterministic LLMs for numerical estimation or hardcoding arbitrary timers, DumHouse features a genuine, mathematically interpretable **Ridge Regression (L2-Regularized Linear Regression)** model written from first principles in TypeScript

(`src/lib/ai/prep-estimator.ts`).

The prediction hypothesis is formulated with standardized z-score features:

```
Predicted Prep Minutes = w0 + SUM [ w_j x (x_j - mean_j) / std_j ]  (for j = 1 to 5)
```

Where x represents the 5 operational kitchen features:

1. **x1 (Item Count):** Total quantity of dishes in the ticket.
2. **x2 (Complexity Points):** Sum of recipe complexity scores on a 1-to-5 scale (for example, "Double Ka Meetha" has complexity 1, whereas slow hand-pounded "Mutton Haleem" has complexity 5).
3. **x3 (Queue Depth):** Number of concurrent active orders currently in `RECEIVED`, `ACCEPTED`, or `PREPARING` stages at that specific outlet.
4. **x4 (Peak Period):** Binary rush indicator (1 during lunch 12:00-15:00 and dinner 19:00-22:00; 0 otherwise).
5. **x5 (Weekend Indicator):** Binary indicator (1 for Friday, Saturday, and Sunday dinner rushes; 0 otherwise).

5.2 Analytical Ridge Closed-Form Solution

The model computes regression weights deterministically via the penalized normal equations without stochastic gradient variance:

```
Weights: w = inv( X^T . X + lambda . I' ) . X^T . y
```

Where regularization parameter `lambda = 1.0`, and `I'` is the identity matrix with `I'[0,0] = 0` (leaving the bias term unpenalized). The matrix inversion is computed using Gauss-Jordan elimination with partial numerical pivoting.

5.3 Model Safeguards & Honest Promises

• **Honest Floor Safeguard:** Promised Time = MAX(Outlet Minimum Prep Time, Predicted Time + ceil(MAE / 2)). The customer promised time is never lower than the outlet's minimum preparation floor.

• **Capacity Fallback:** If ticket item count exceeds 50 or kitchen queue exceeds 40 active handis, the system switches to a conservative catering bulk heuristic (1.5x buffer) and flags the ticket.

• **Staff Override with Audit Log:** Kitchen leads can manually adjust prep times directly from the KDS screen with mandatory reason logging (`CHEF_ADJUSTMENT`, `RUSH_HOUR`, `EQUIPMENT_MAINTENANCE`).

• **Late Order Escalation (>10m Alert):** If an order's elapsed time exceeds the promised time by 10 minutes or more, the KDS pulses a high-visibility crimson alert and dispatches customer notification updates.

• **Non-Intrusive Reorders:** AI reorder suggestions from customer order history provide quick-action prompts but never inject items into the cart without explicit customer tap.

6. User Interface & Design Decisions

6.1 Design Philosophy: Nizami Hospitality Meets Operational Clarity

• **Palette:** Warm terracotta (`#D9531E`), Hyderabadi saffron amber (`#EAB308`), rich charcoal typography (`#1C1917`), and natural cream surfaces (`#FAF8F5`).

• **Customer Mobile Experience:** Designed for fast, hungry scanning with large food photography, dietary symbols, allergen chips, and one-tap reorders.

• **Kitchen Tablet KDS:** High-contrast 2-meter readable interface, large ticket numbers, and color-coded late warnings.

7. Security, Privacy & Regulatory Compliance

1. **Digital Personal Data Protection (DPDP) Act 2023:**

• Minimal customer data collection with explicit opt-in for marketing messages at checkout.

• Strict role-based isolation preventing customers from accessing unauthorized order records.

2. **Statutory 5% GST Disclosure:**

• All displayed prices across menu, cart, and checkout include 5% statutory restaurant GST.

3. **FSSAI Food Hygiene Licensing:**

• Outlets display statutory FSSAI numbers with sample/demo compliance disclaimers.

8. Verification & Automated Test Results

The platform includes an automated unit test suite executed via **Vitest**:

```
[PASS] tests/ai-prep-estimator.test.ts (6 tests)
  [PASS] trains successfully on synthetic restaurant kitchen dataset
  [PASS] yields positive weights for items, complexity and queue depth
  [PASS] enforces outlet minimum preparation time safeguard
  [PASS] uses conservative statistical fallback when model has insufficient data
  [PASS] uses capacity fallback for extreme party orders
  [PASS] correctly identifies confidence level based on feature distribution

[PASS] tests/order-business-logic.test.ts (8 tests)
  [PASS] allows all valid forward status transitions
  [PASS] permits cancellation from early stages but not terminal states
  [PASS] strictly rejects backward and invalid jump transitions
  [PASS] calculates order prices securely server-side from menu snapshot
  [PASS] correctly applies coupon code discount and loyalty point deductions
  [PASS] accurately applies add-on side costs to line items
  [PASS] handles kitchen staff prep time override with audit trail
  [PASS] prevents customer from querying or modifying other customer private data

Test Files: 2 passed (2) | Tests: 14 passed (14)
```

9. Explicit Scope Reductions (What Was Cut)

1. **Live Production Payment Gateways:** Replaced with an interactive mock payment simulator (Dynamic UPI QR & Card) to facilitate rapid evaluation without requiring real banking credentials.
2. **Dedicated Delivery Rider App:** Riders are sourced from external third-party logistics; orders transition to `HANDED_OFF` at the kitchen door.
3. **Replacing Outlet Billing (POS) or Inventory:** Handled via direct menu availability toggling and KDS.
4. **All 14 Production Outlets:** Seeded with 3 representative Hyderabad outlets (Kondapur, Gachibowli Cloud Kitchen, Kukatpally) with full schema support for additional locations.

10. Labeled Assumptions Table

Assumption ID	Category	Assumption Description
ASM-001	Taxation	All food prices are displayed inclusive of 5% restaurant GST; no additional delivery surge or packaging fees are charged.
ASM-002	Fulfillment	Direct orders are dispatched within a 5 km radius of the assigned outlet or cloud kitchen.
ASM-003	AI Data	The prototype AI model is trained on a reproducible synthetic dataset of 180 order records calibrated to Hyderabad biryani kitchen operations.
ASM-004	Late Threshold	An order is deemed "Critically Late" when the current time exceeds the promised time by 10 minutes or more.
ASM-005	Party Notice	Party catering orders require a mandatory minimum 24-hour advance notice to allow charcoal dum bhatti preparation.
ASM-006	Demo Access	Demo credentials (`customer.demo@example.com`, `kitchen.demo@example.com`) are deliberately public for assessment reviewers.